

```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [3]: df=pd.read_csv("Telco-Customer-Churn.csv")
df.head(5)
```

Out[3]:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	Mul
--	------------	--------	---------------	---------	------------	--------	--------------	-----

0	7590-VHVEG	Female	0	Yes	No	1	No	
---	------------	--------	---	-----	----	---	----	--

1	5575-GNVDE	Male	0	No	No	34	Yes	
---	------------	------	---	----	----	----	-----	--

2	3668-QPYBK	Male	0	No	No	2	Yes	
---	------------	------	---	----	----	---	-----	--

3	7795-CFOCW	Male	0	No	No	45	No	
---	------------	------	---	----	----	----	----	--

4	9237-HQITU	Female	0	No	No	2	Yes	
---	------------	--------	---	----	----	---	-----	--

5 rows × 21 columns



```
In [5]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customerID            7043 non-null   object
1   gender                 7043 non-null   object
2   SeniorCitizen          7043 non-null   int64
3   Partner                7043 non-null   object
4   Dependents             7043 non-null   object
5   tenure                 7043 non-null   int64
6   PhoneService           7043 non-null   object
7   MultipleLines          7043 non-null   object
8   InternetService        7043 non-null   object
9   OnlineSecurity         7043 non-null   object
10  OnlineBackup           7043 non-null   object
11  DeviceProtection       7043 non-null   object
12  TechSupport            7043 non-null   object
13  StreamingTV            7043 non-null   object
14  StreamingMovies        7043 non-null   object
15  Contract               7043 non-null   object
16  PaperlessBilling       7043 non-null   object
17  PaymentMethod          7043 non-null   object
18  MonthlyCharges         7043 non-null   float64
19  TotalCharges           7043 non-null   object
20  Churn                  7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

```
In [6]: df.isnull().sum()
```

```
Out[6]: customerID      0
gender                0
SeniorCitizen         0
Partner               0
Dependents            0
tenure                0
PhoneService          0
MultipleLines         0
InternetService       0
OnlineSecurity        0
OnlineBackup          0
DeviceProtection      0
TechSupport           0
StreamingTV           0
StreamingMovies       0
Contract              0
PaperlessBilling      0
PaymentMethod         0
MonthlyCharges        0
TotalCharges          0
Churn                 0
dtype: int64
```

```
In [7]: df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")
```

```
In [ ]: #Converted TotalCharges from object datatype to numeric for accurate revenue cal
```

```
In [8]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customerID            7043 non-null   object
1   gender                 7043 non-null   object
2   SeniorCitizen          7043 non-null   int64
3   Partner                7043 non-null   object
4   Dependents             7043 non-null   object
5   tenure                 7043 non-null   int64
6   PhoneService           7043 non-null   object
7   MultipleLines          7043 non-null   object
8   InternetService        7043 non-null   object
9   OnlineSecurity         7043 non-null   object
10  OnlineBackup           7043 non-null   object
11  DeviceProtection       7043 non-null   object
12  TechSupport            7043 non-null   object
13  StreamingTV            7043 non-null   object
14  StreamingMovies        7043 non-null   object
15  Contract               7043 non-null   object
16  PaperlessBilling       7043 non-null   object
17  PaymentMethod          7043 non-null   object
18  MonthlyCharges         7043 non-null   float64
19  TotalCharges           7032 non-null   float64
20  Churn                  7043 non-null   object
dtypes: float64(2), int64(2), object(17)
memory usage: 1.1+ MB
```

```
In [8]: #Dataset contained no missing values after validation.
```

```
In [9]: # Business Problem
```

```
#Customer churn occurs when customers stop using telecom services.
```

```
Cell In[9], line 3
```

```
Customer churn occurs when customers stop using telecom services.
```

```
^
```

```
SyntaxError: invalid syntax
```

```
In [9]: churn_counts = df["Churn"].value_counts()
print(churn_counts)
```

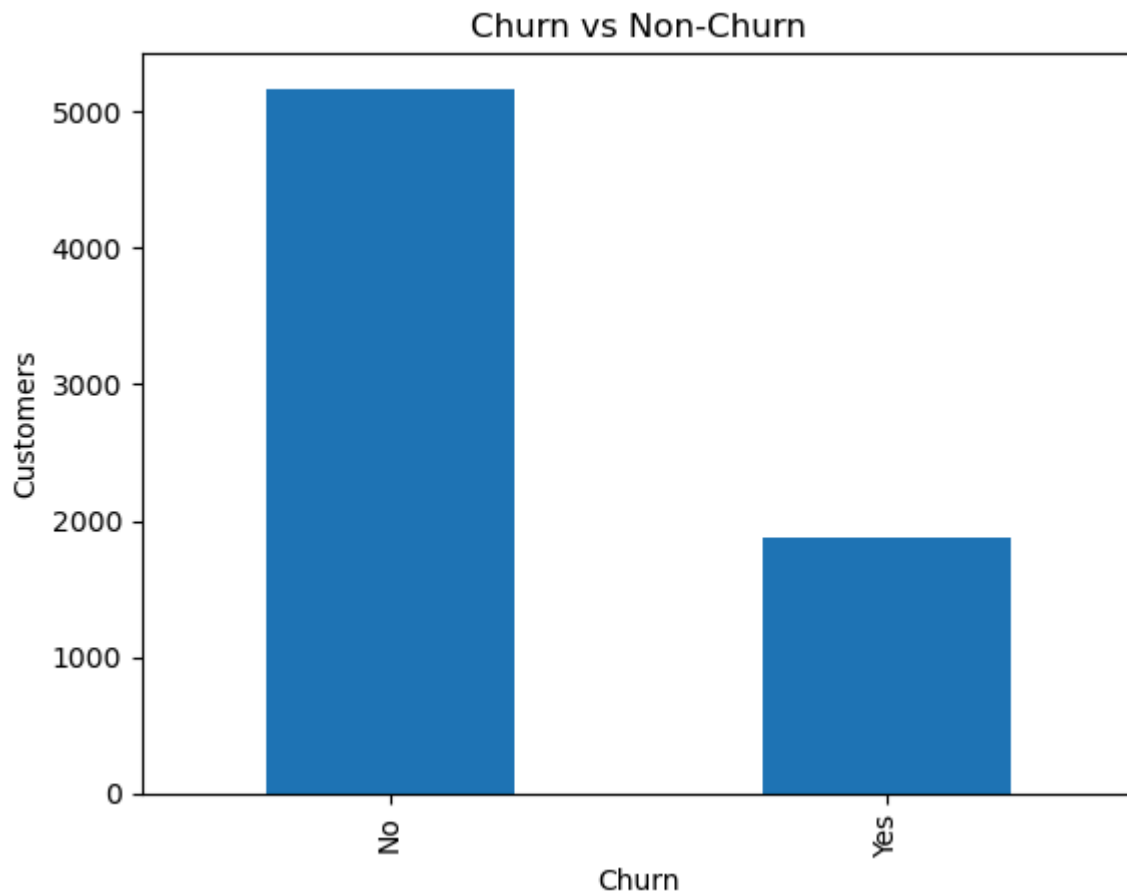
```
Churn
```

```
No      5174
```

```
Yes     1869
```

```
Name: count, dtype: int64
```

```
In [10]: churn_counts.plot(kind="bar")
plt.title("Churn vs Non-Churn")
plt.xlabel("Churn")
plt.ylabel("Customers")
plt.show()
```

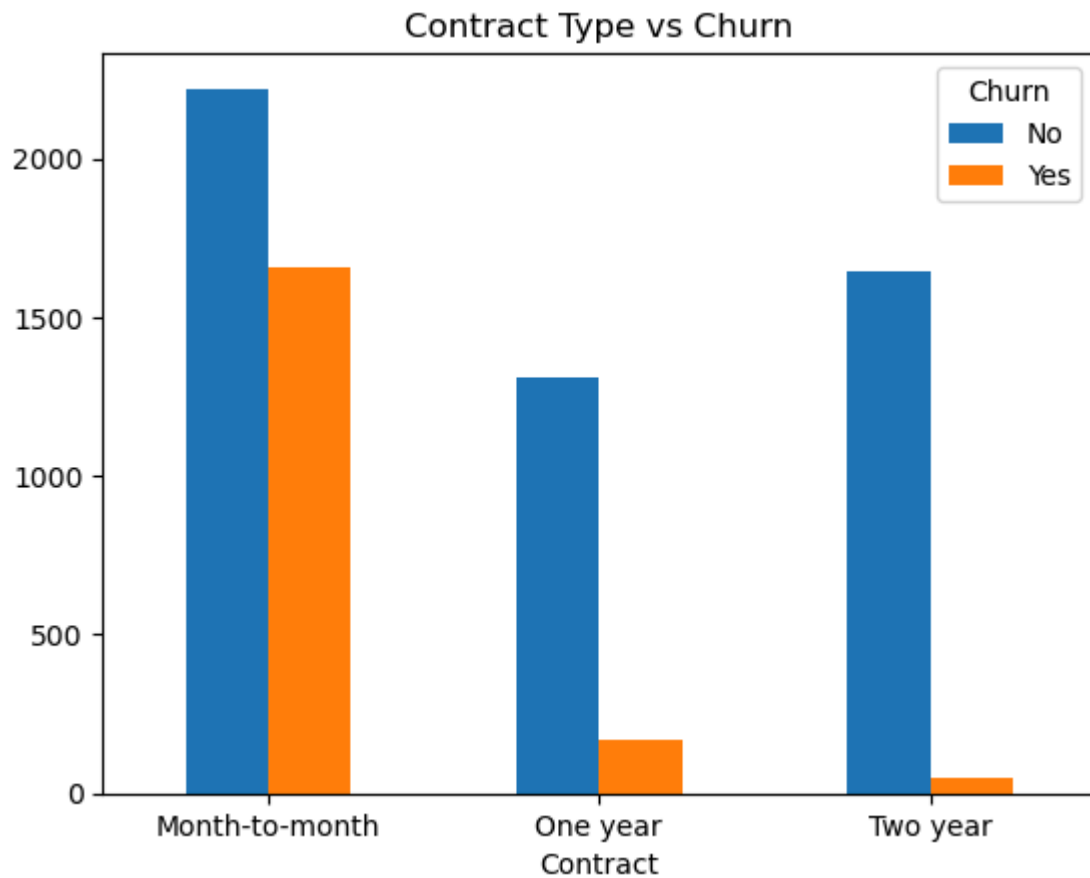


```
In [11]: contract_churn = pd.crosstab(df["Contract"], df["Churn"])

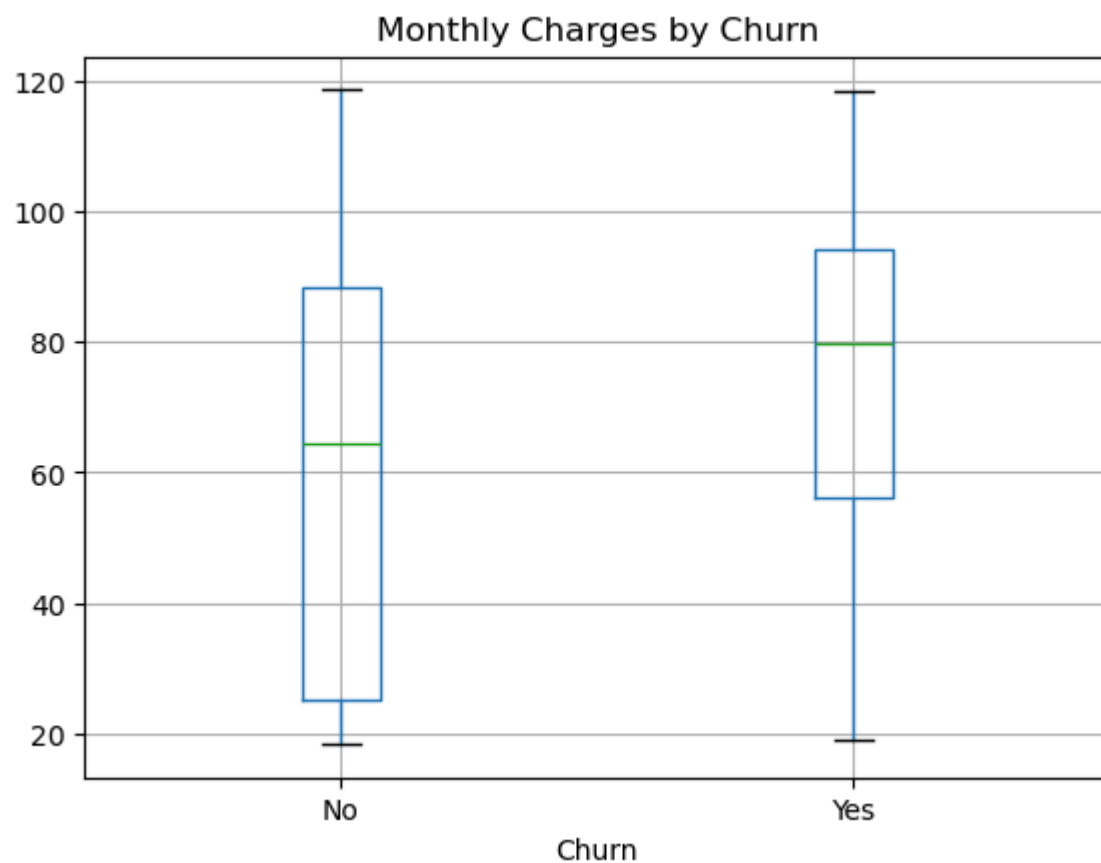
print(contract_churn)

contract_churn.plot(kind="bar")
plt.title("Contract Type vs Churn")
plt.xticks(rotation=0)
plt.show()
```

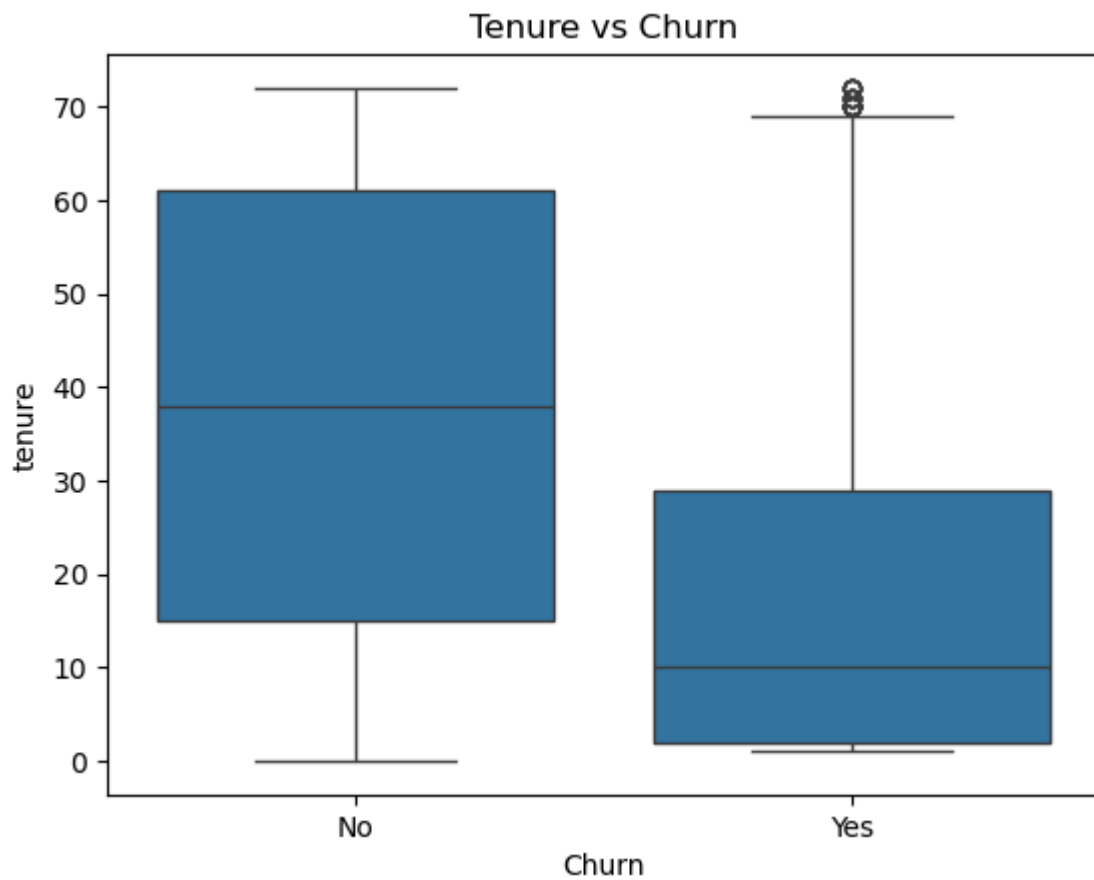
Churn	No	Yes
Contract		
Month-to-month	2220	1655
One year	1307	166
Two year	1647	48



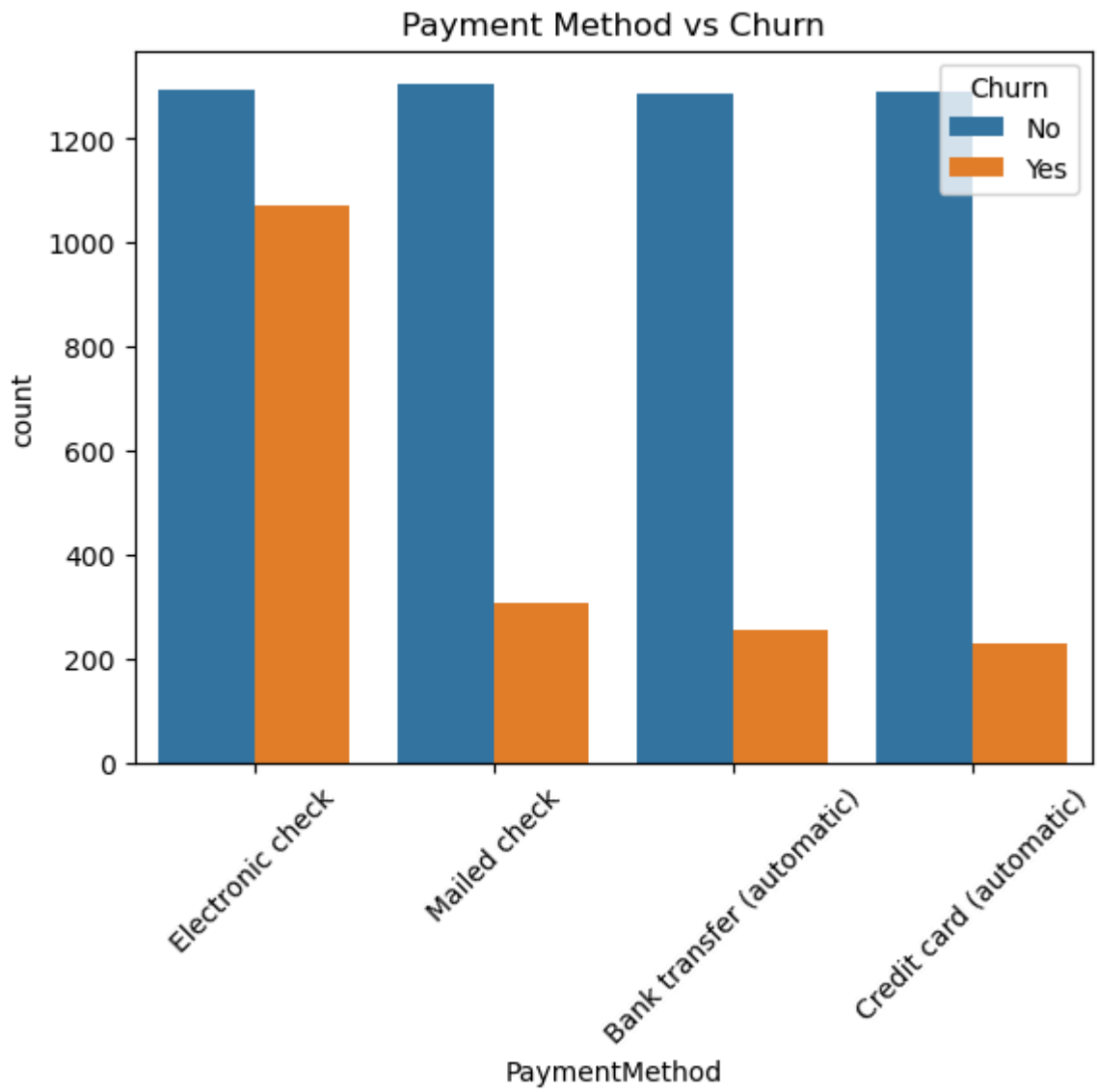
```
In [12]: df.boxplot(column="MonthlyCharges", by="Churn")  
plt.title("Monthly Charges by Churn")  
plt.suptitle("")  
plt.show()
```



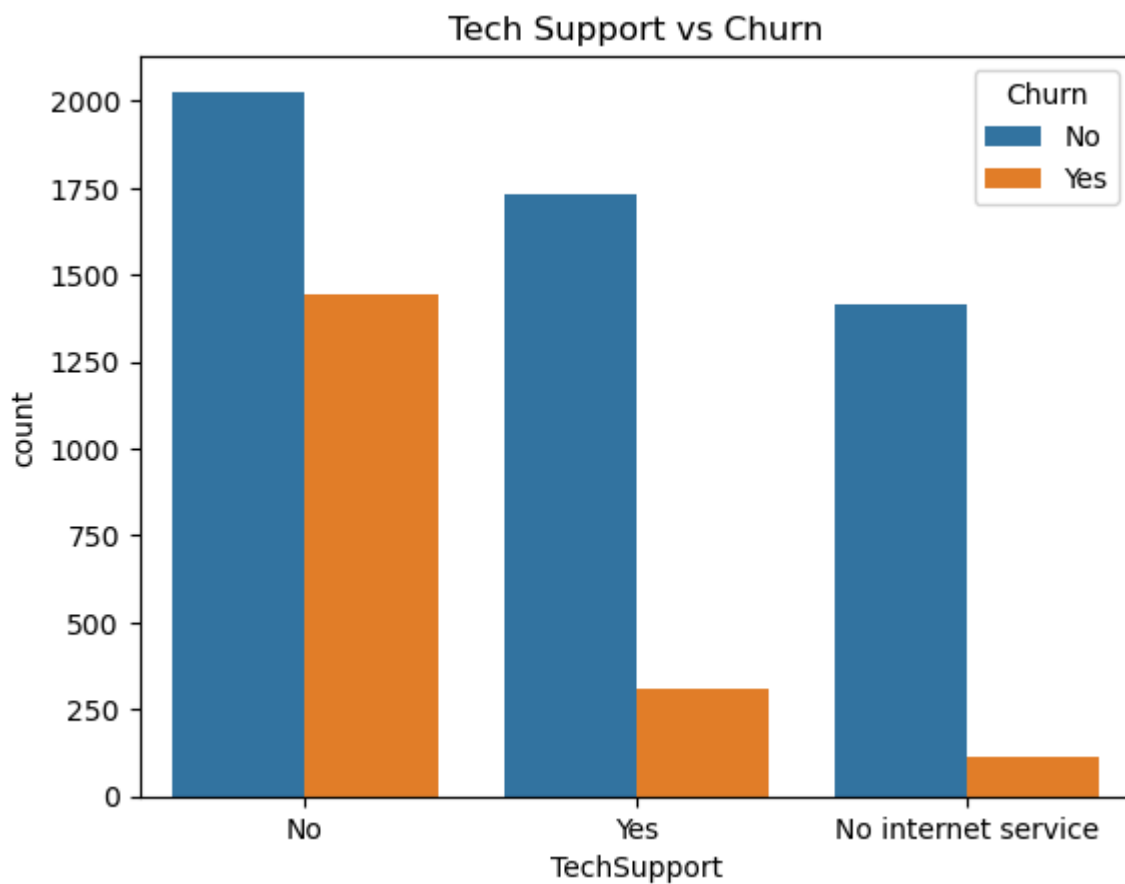
```
In [13]: sns.boxplot(x="Churn", y="tenure", data=df)
plt.title("Tenure vs Churn")
plt.show()
```



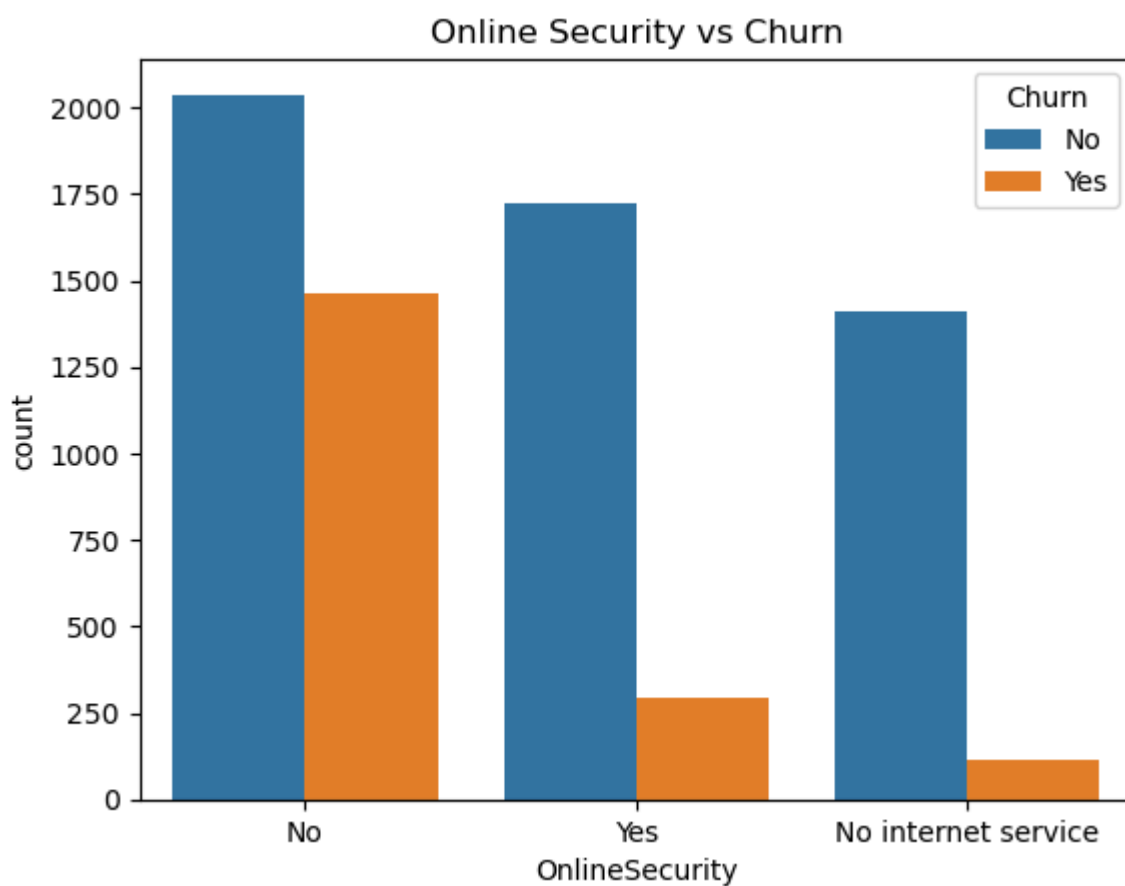
```
In [16]: sns.countplot(x="PaymentMethod", hue="Churn", data=df)
plt.xticks(rotation=45)
plt.title("Payment Method vs Churn")
plt.show()
```



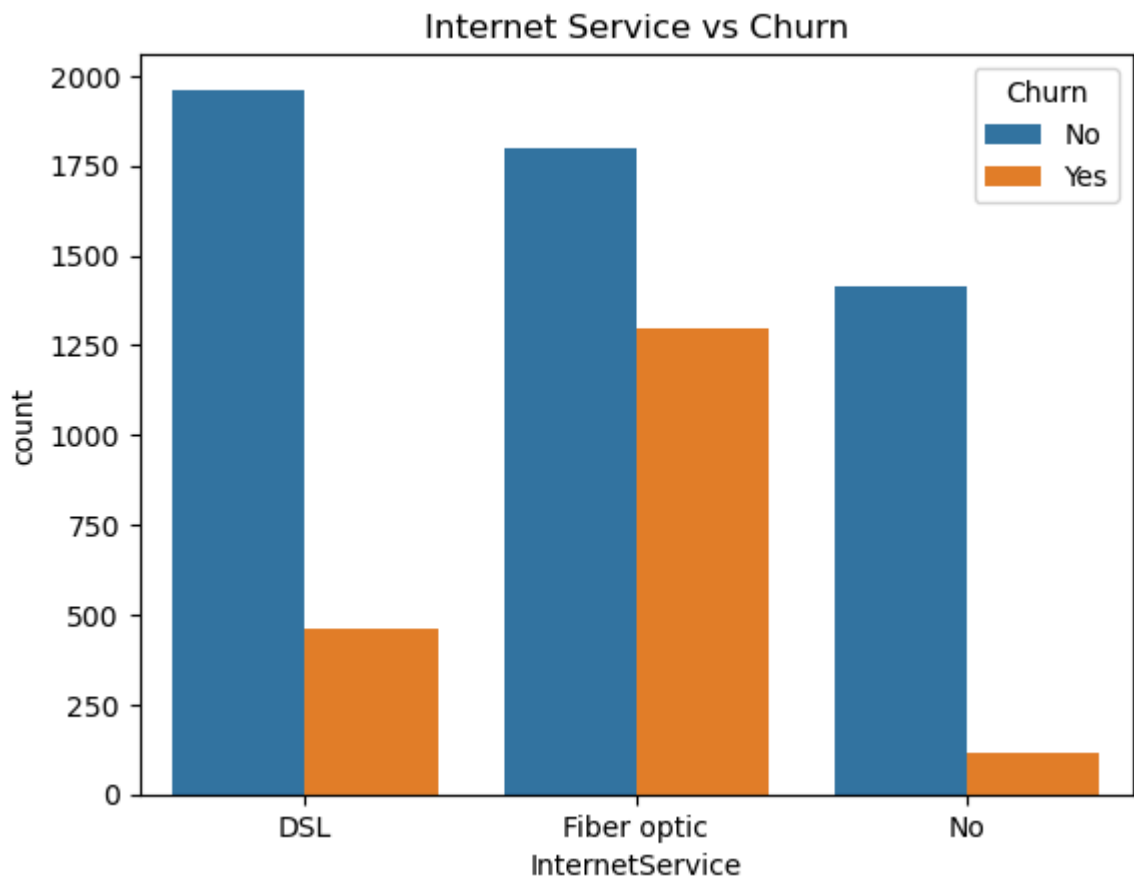
```
In [17]: sns.countplot(x="TechSupport", hue="Churn", data=df)
plt.title("Tech Support vs Churn")
plt.show()
```



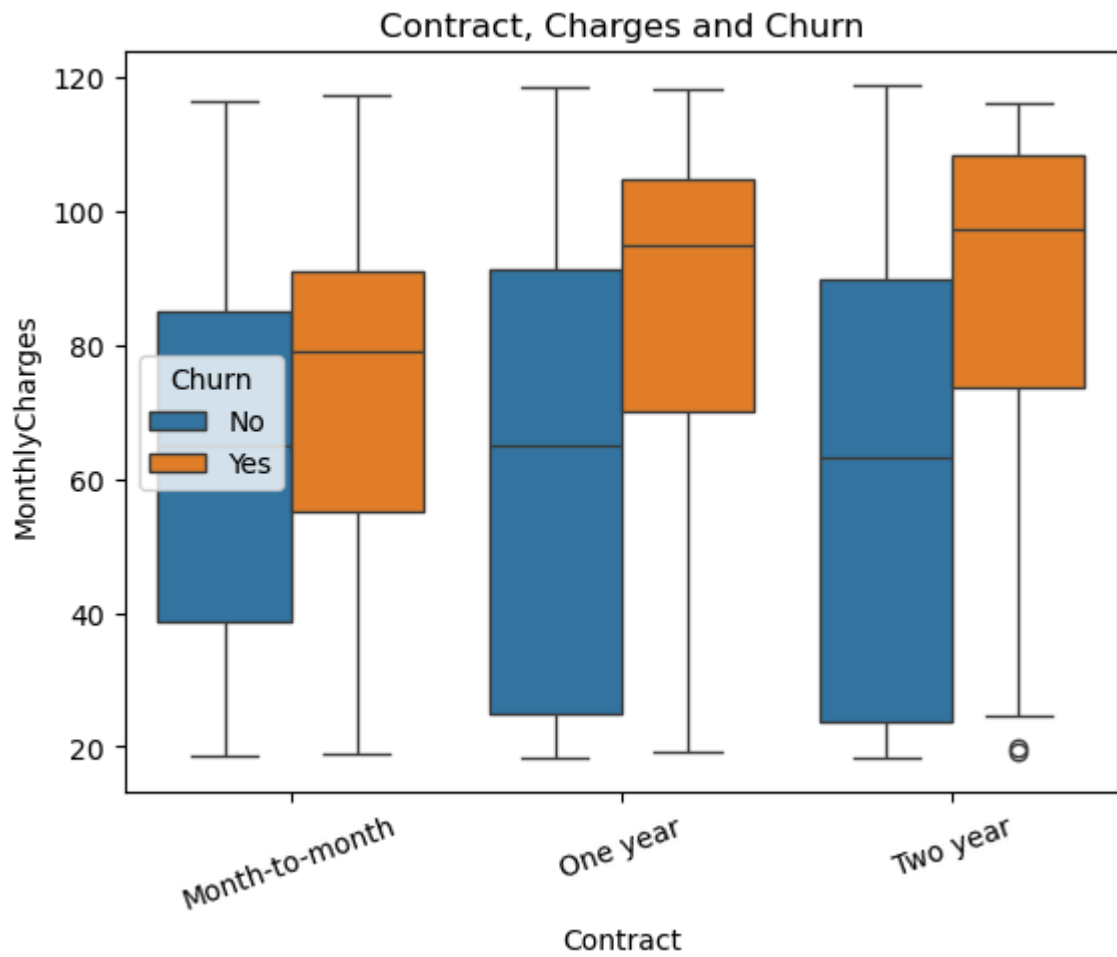
```
In [18]: sns.countplot(x="OnlineSecurity", hue="Churn", data=df)
plt.title("Online Security vs Churn")
plt.show()
```



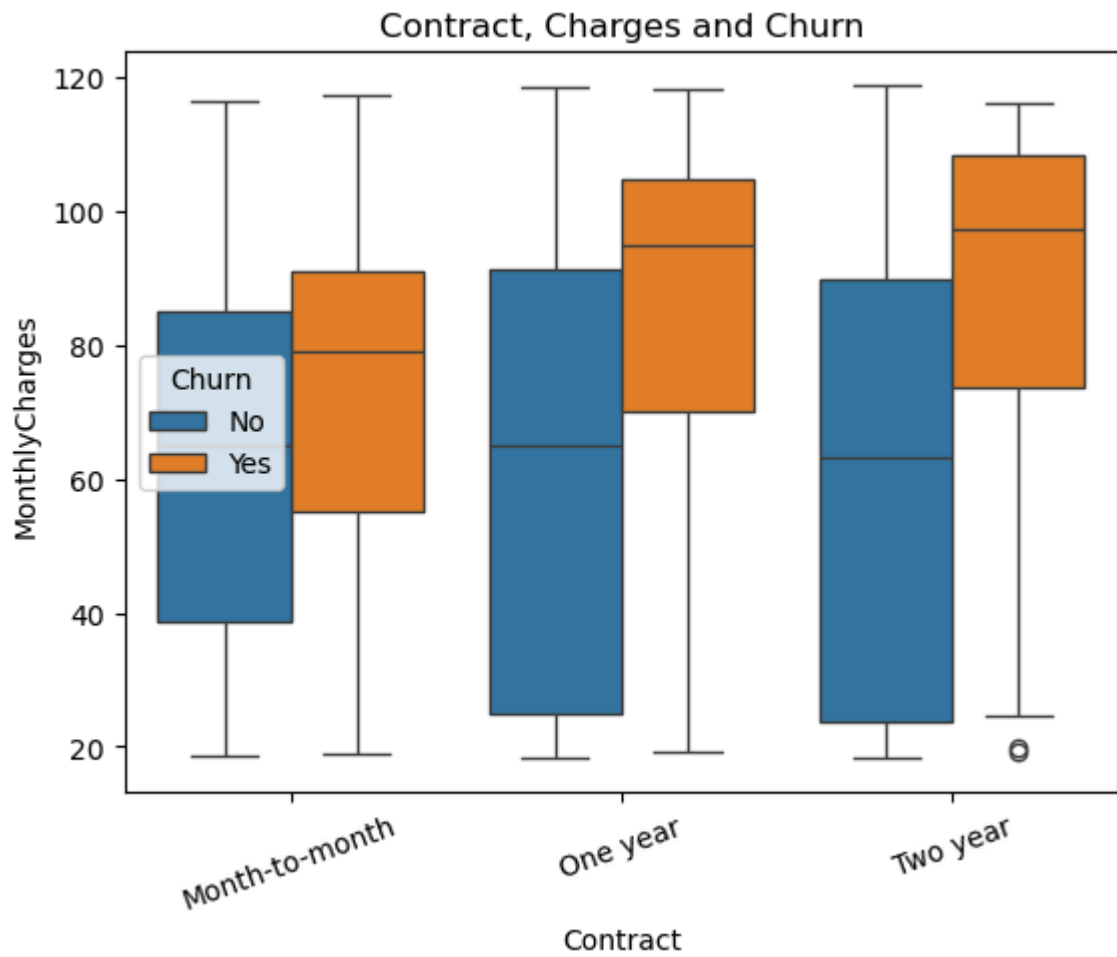

```
In [19]: sns.countplot(x="InternetService", hue="Churn", data=df)
plt.title("Internet Service vs Churn")
plt.show()
```



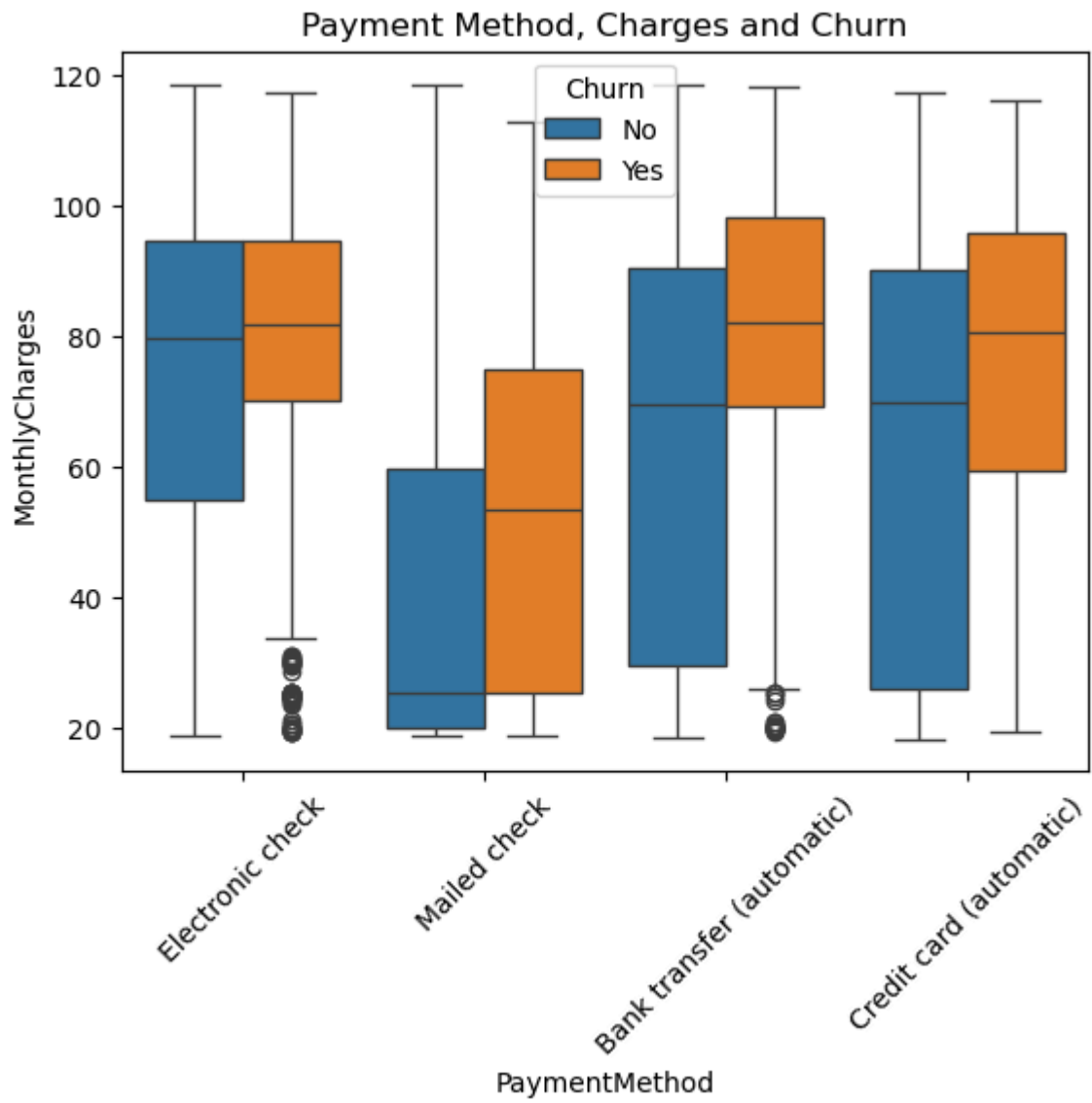
```
In [20]: sns.boxplot(
    x="Contract",
    y="MonthlyCharges",
    hue="Churn",
    data=df
)
plt.xticks(rotation=20)
plt.title("Contract, Charges and Churn")
plt.show()
```



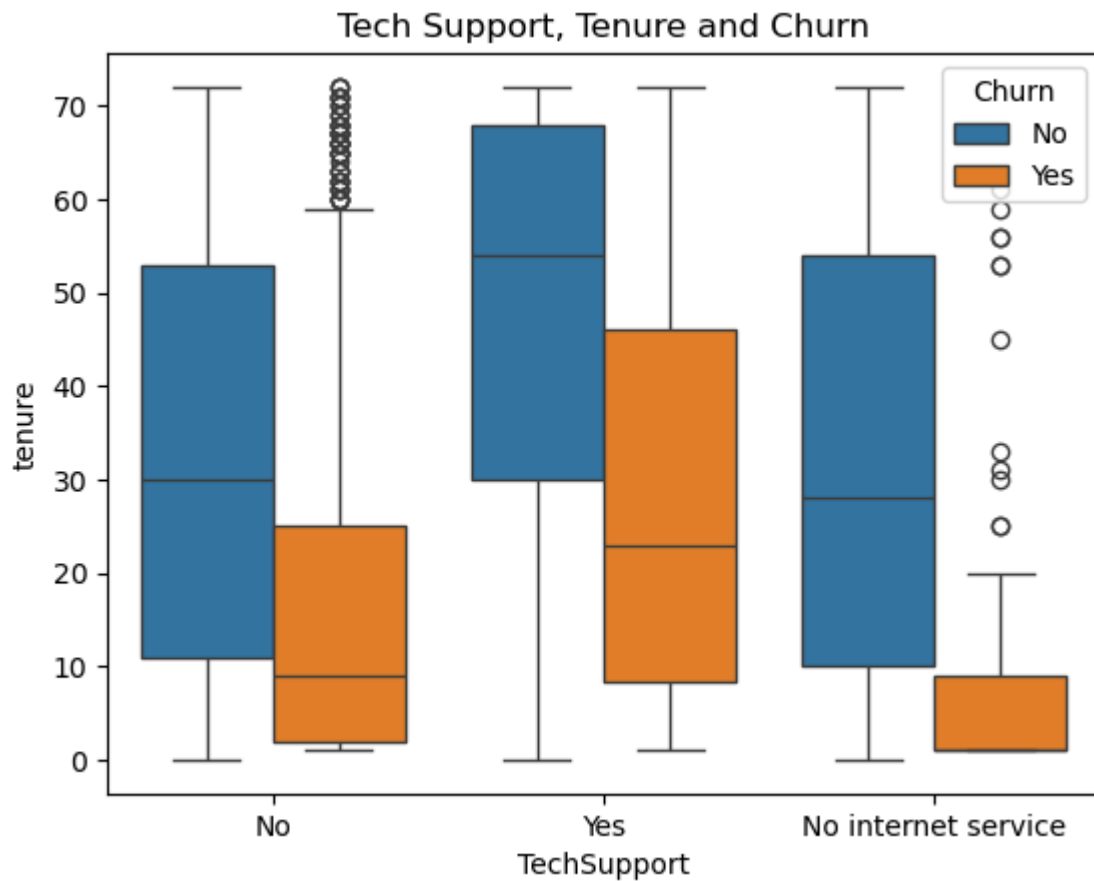
```
In [21]: sns.boxplot(  
    x="Contract",  
    y="MonthlyCharges",  
    hue="Churn",  
    data=df  
)  
plt.xticks(rotation=20)  
plt.title("Contract, Charges and Churn")  
plt.show()
```



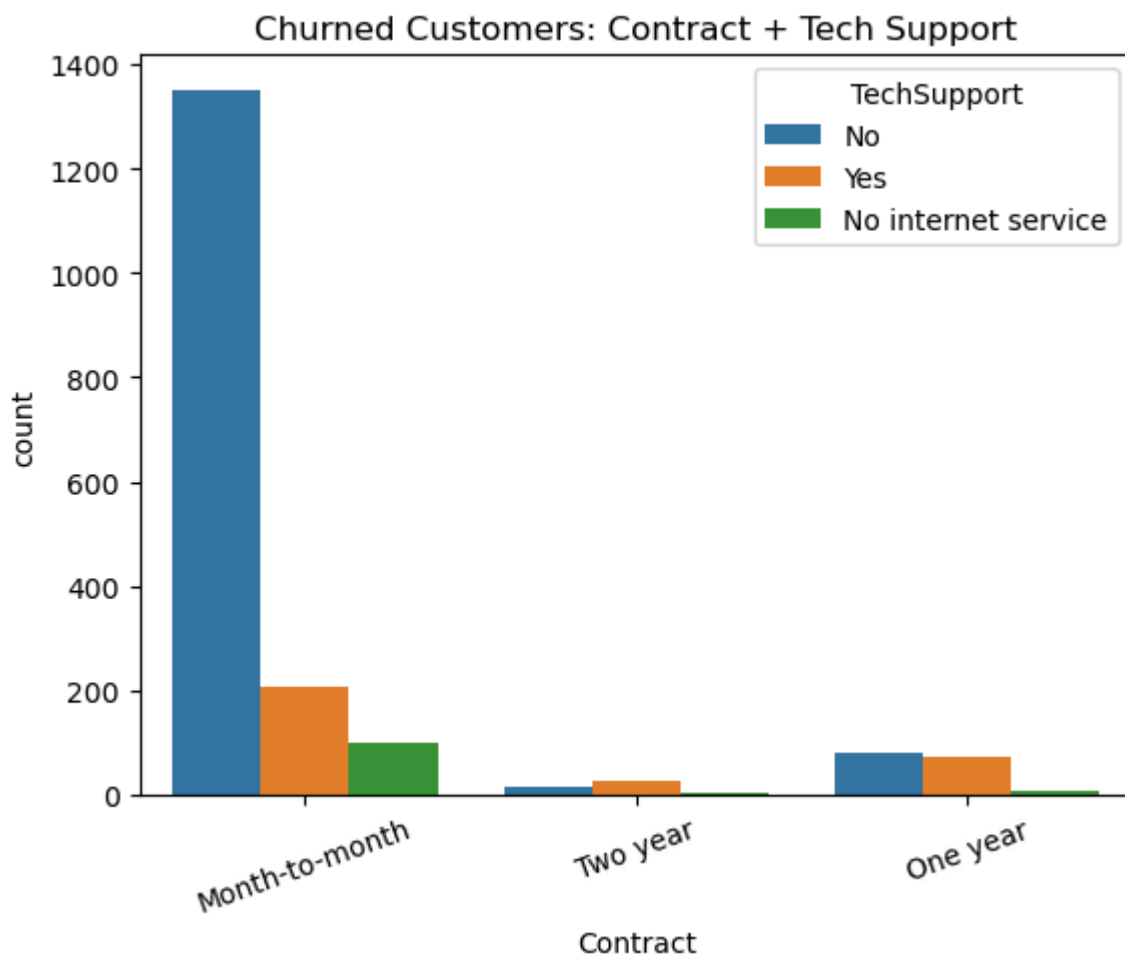
```
In [22]: sns.boxplot(  
    x="PaymentMethod",  
    y="MonthlyCharges",  
    hue="Churn",  
    data=df  
)  
plt.xticks(rotation=45)  
plt.title("Payment Method, Charges and Churn")  
plt.show()
```



```
In [23]: sns.boxplot(  
    x="TechSupport",  
    y="tenure",  
    hue="Churn",  
    data=df  
)  
plt.title("Tech Support, Tenure and Churn")  
plt.show()
```



```
In [24]: sns.countplot(  
    x="Contract",  
    hue="TechSupport",  
    data=df[df["Churn"]=="Yes"]  
)  
plt.xticks(rotation=20)  
plt.title("Churned Customers: Contract + Tech Support")  
plt.show()
```



```
In [15]: loyal = df[
    (df["Contract"]!="Month-to-month") &
    (df["tenure"]>24)
]

print("Loyal Customers:",len(loyal))
```

Loyal Customers: 2689

```
In [14]: high_risk = df[
    (df["Contract"]=="Month-to-month") &
    (df["MonthlyCharges"]>70) &
    (df["tenure"]<12)
]

print("High Risk:",len(high_risk))
```

High Risk: 814